Paleomagnetic study of the Late Ordovician – Early Silurian platform sequence of Anticosti Island, Quebec[1]

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Eighteen sites (152 samples, 441 specimens) from the Late Ordovician – Early Silurian sequence of Anticosti Island were studied paleomagnetically. Six sites were collected in the Vauréal Formation, seven in the Ellis Bay Formation, three in the Becscie Formation, and one each in the Gun River and Jupiter formations. The lithologies sampled consisted of limestones, sandy limestones, marls, and sandy calcareous shales drawn from a sedimentary platform sequence of predominantly interbedded limestones and shales. The specimens were demagnetized in alternating fields (AF) and thermally and were found to be quite stable. Two components of magnetization were isolated. The memory carrier is fine- to medium-grained magnetite; AF and thermal cleanings are about equally efficient. The average directions of residual magnetization are $D = 167°$, $I = 37°$, $\alpha_{95} = 18°$; $D = 315°$, $I = -24°$, $\alpha_{95} = 17°$ for components B and C, respectively. The corresponding paleopoles for the B and C components are 129°E, 19°N (dp = 12°, dm = 21°) and 341°E, 16°S (dp = 10°, dm = 18°).

Component A is most probably composed of component C and the present Earth's field (PEF); it was chiefly observed in the more altered Vauréal Formation and in the altered top part of the cored samples. The most common component in the Ellis Bay, Becscie, Gun River, and Jupiter formations is component C. Component B is mainly found in the Becscie and Ellis Bay formations. Both components B and C are isolated in the 300–500°C and 10–40 mT ranges. Component C is either older than component B or synchronous and of reverse polarity to component B. If this last alternative is correct, then the Ordovician and Silurian formations are overprinted by a pre-Kiaman magnetization. The more plausible scenario is the following: component B is secondary, of pre-Middle Carboniferous age, and overprinted on an older secondary magnetization (component C), the acquisition age of which is confined to the Silurian–Carboniferous time interval.

Une étude paléomagnétique a porté sur 152 échantillons (441 spécimens) recueillis à 18 emplacements différents dans la séquence de l'Ordovicien supérieur – Silurien inférieur de l'île d'Anticosti. Six emplacements de la formation de Vauréal ont été échantillonnés, sept dans la formation d'Ellis Bay, trois dans la formation de Becscie, et un dans chacune des formations de Gun River et de Jupiter. Les unités lithologiques échantillonnées sont formées de calcaires, de calcaires sableux, de marnes et de schistes argileux calcaires et sableux provenant d'une séquence sédimentaire de plate-forme dominée par des calcaires et des schistes argileux interstratifiés. Les méthodes de champs alternatifs (CA) et de traitement thermique ont été utilisées pour désaimanter les échantillons, et on a observé que leur comportement était relativement stable. Deux composantes d'aimantation ont été reconnues. La mémoire magnétique est portée par la magnétite de grain fin à moyen; les désaimantations par CA ou par traitement thermique donnent sensiblement le même rendement. Les directions moyennes d'aimantation résiduelle pour les composantes B et C sont $D = 167°$, $I = 37°$, $\alpha_{95} = 18°$; $D = 315°$, $I = -24°$, $\alpha_{95} = 17°$, respectivement. Les paléopôles qui correspondent aux composantes B et C sont 129°E., 19°N. (dp = 12°, dm = 21°) et 341°E., 16°S. (dp = 10°, dm = 18°).

La composante A comprend possiblement la composante C et le champ magnétique actuel (CMA); elle fut observée pricipalement dans la partie la plus altérée de la formation de Vauréal et dans la partie supérieure altérée des échantillons prélevés par forage. La composante C domine dans les formations d'Ellis Bay, Becscie, Gun River et Jupiter. La composante B est observée surtout dans les formations de Becscie et d'Ellis Bay. Les deux composantes B et C deviennent isolées entre 300 et 500°C et entre 10 et 40 mT. La composante C est plus ancienne que la composante B ou synchrone et de polarité inversée à la composante B. Si cette dernière interprétation est juste, les formations de l'Ordovicien et du Silurien sont surimpressionnées par une aimantation anté-kiamienne. Le scénario le plus acceptable est le suivant : la composante B est secondaire, elle est d'âge anté-carbonifère moyen et elle est surimpressionnée sur une aimantation secondaire plus ancienne (composante C) et l'âge d'acquisition serait restreint à la durée de l'intervalle du Silurien–Carbonifère.

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Introduction

The essentially horizontal and undisturbed Late Ordovician to Early Silurian platform succession of Anticosti Island in the Gulf of St. Lawrence (Fig. 1) provides an ideal laboratory for paleomagnetic studies. It represents the Cambrian–Ordovician passive margin of the Proto-Atlantic Ocean. A preliminary paleomagnetic study of the lower part of this sequence by Seguin and Petryk (1984) showed that the polarity of remanence is normal in the Vauréal and the lower Ellis Bay formations (Late Ordovician), mainly reversed in the upper Ellis Bay formation (Late Ordovician), and normal in the Becscie Formation (Early Silurian). It was suggested that the apparent polar wandering path (APWP) across the Ordovician–Silurian boundary in North America may be appreciably different from the accepted APWP model for this time interval (Van der Voo 1981). Furthermore, the preliminary results (Seguin and Petryk 1984) indicated that if the remanence is primary, little APW took place during Late Ordovician through Early Silurian time. The paleolatitude

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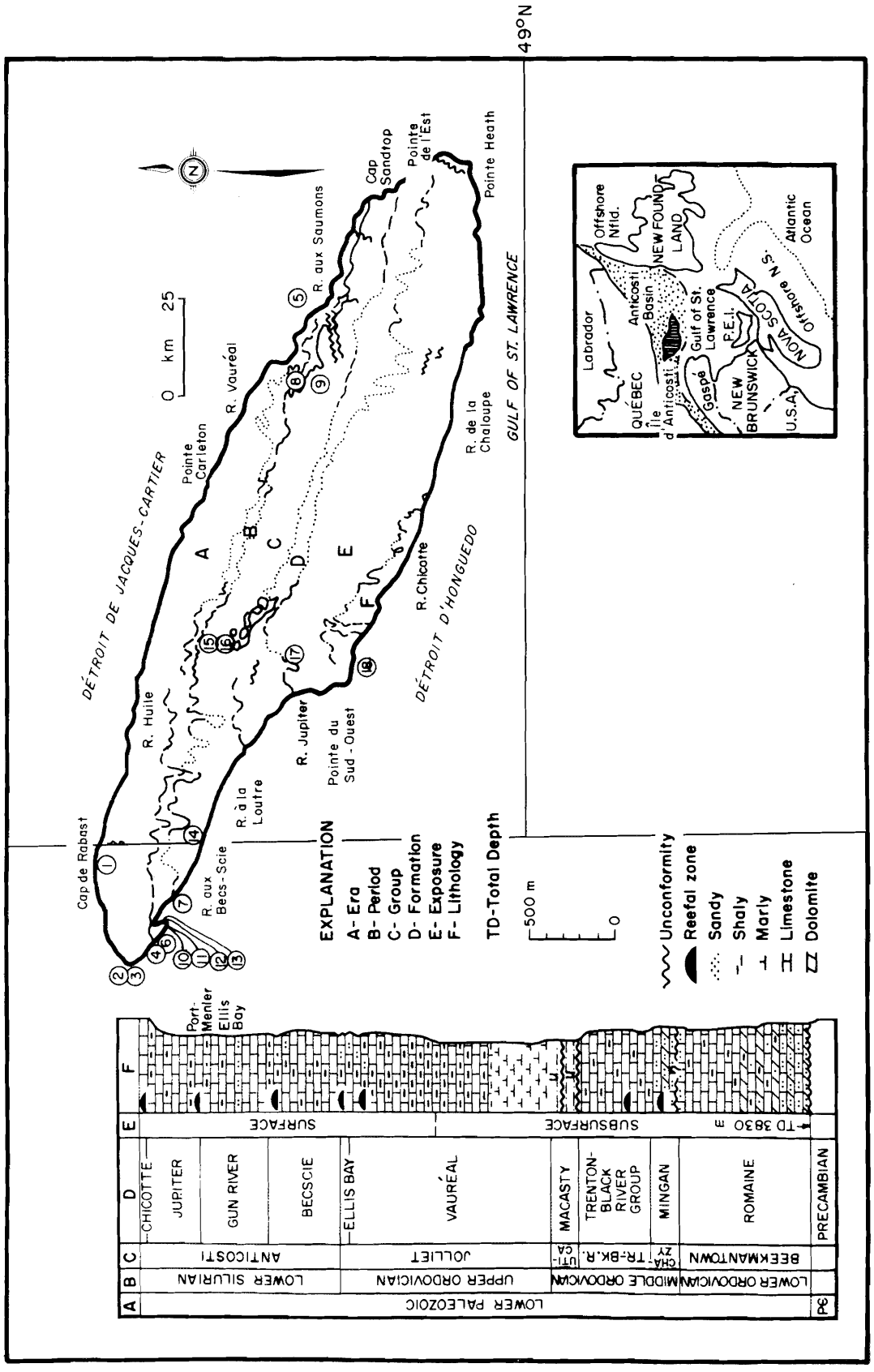


FIG. 1. Generalized geological map of Anticosti Island, stratigraphic column, locations of sampling sites.

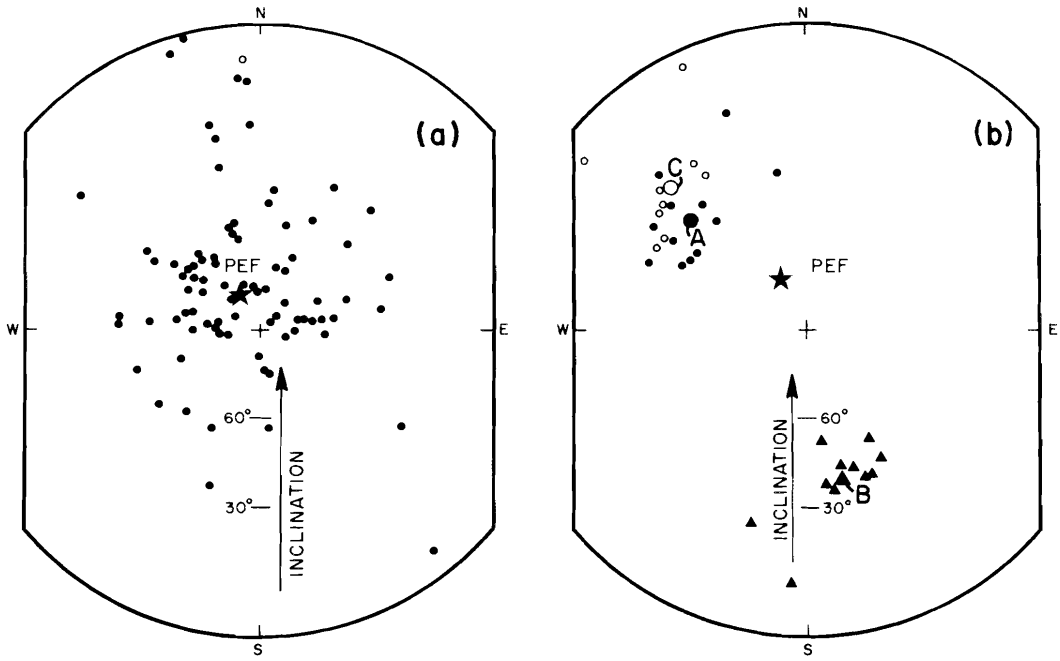


FIG. 2. (*a*) Natural remanent directions plotted on Wulff net for these Upper Ordovician and Lower Silurian sedimentary rock units. Solid circles (open circles) represent projections on lower (upper) hemisphere. The star represents the PEF in the sampling region. (*b*) Site-mean directions of magnetization obtained after treatment plotted on Wulff net. Solid circles represent component A; triangles, component B; and open circles, component C. The mean directions of components A, B, and C are indicated by the larger solid circle, triangle, and open circle, respectively.

remained roughly the same for this time interval, with Anticosti located slightly north of the paleoequator.

The purpose of this second and more elaborate paleomagnetic study is to verify the validity of the conclusions obtained in our preliminary study and to obtain an embryonic magnetostratigraphic column of the Late Ordovician – Early Silurian boundary succession of North America. Paleomagnetic results of lower Paleozoic rock units are very difficult to interpret because of the absence of a reliable APWP. This lack of an APWP is mainly attributable to the fact that the remanence acquisition age of lower Paleozoic rock units is difficult to pinpoint because of extensive overprinting due to the occurrence of several orogenic episodes: Taconic, Acadian, and Hercynian.

Geological setting

The roughly 9000 km² Anticosti Island constitutes the eastern segment of the physiographic region of the St. Lawrence Platform. It lies between the Canadian Shield to the north and the Appalachian Orogen to the south and east. The absence of significant deformational structures in the Anticosti sequence demonstrates that the Anticosti Paleoplatform remained stable through at least two tectonic upheavals: the Taconic and Acadian orogenies. Consequently, old (eventually primary and free from overprints) remanences can be expected on this platform. Stratal deformations attributed to these orogenic events are mainly restricted to the south of the ancient geosuture called Logan's Line (Poole *et al.* 1969). The basal Anticosti stratigraphic sequence rests unconformably on the southeast margin of the Precambrian craton (Richardson 1857). The platform sequence, which forms a wedge from north to south, thickens to in excess of 3800 m south of Anticosti Island. Carbonates, marls, shales, and siliciclastic sands, possibly Cambrian and Early Middle, and in part Late Ordovician in age (Twenhofel 1928; Roliff 1968; Riva 1969), occur mainly as submarine and subsurface units. The succeeding Late Ordovician and Early Silurian age formations occur offshore and south of the island.

The approximately 1100 m sequence of shallow, southwest-dipping (~2°) strata of Anticosti Island consists of six formations within two major groups: the Upper Ordovician Jolliet Group, comprising the Vauréal and Ellis Bay formations; and the mainly Lower Silurian Anticosti Group, comprising the Becscie, Gun River, Jupiter, and Chicotte formations (Petryk 1979, 1981*a*, 1981*b*). Together, these formations constitute the most complete, uninterrupted sequence of Late Ordovician to Early Silurian age strata in eastern North America (Billings 1857; Schuchert and Twenhofel 1910; Twenhofel 1928; Bolton 1972, 1981; Petryk 1981*a*, 1982). Lime carbonates, ranging from lime mudstone to boundstone to intraformational conglomerate interbedded with calcareous shale, predominate in the exposed Anticosti sequence. Although siltstones and sandstones are, in general, rare, silt and sand are widely distributed in the sequence, usually as partings or gradational admixtures between carbonate and shale beds; it is in these lithologies, which are more magnetic, that most of the samples of this study have been derived. The bulk of the siliciclastic silt and sand occurs within the upper Vauréal and the Ellis Bay formations and in the eastern transitional carbonate–siliciclastic facies, predominantly to the east of Saumons River (Rivière aux Saumons). Sand is relatively scarce in the succeeding formations.

As a result of sedimentologic, biostratigraphic, and sequence analyses, Petryk (1979, 1981*b*) has evoked a major megacycle model of continuous sedimentation for the Anticosti Platform during Ashgill and Llandovery time. The "turn-

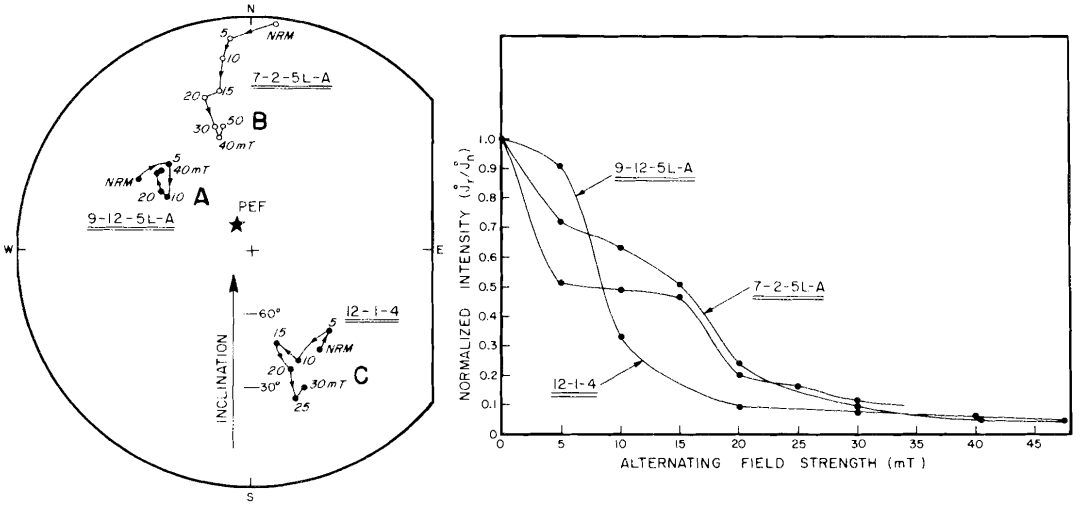


FIG. 3. Typical A, B, and C directions and intensities obtained by AF demagnetization of specimens from low- and intermediate-intensity core. Intensities are normalized to the NRM values. Directions are plotted on a Wulff projection with closed (open) symbols indicating downward (upward) vectors. Alternating fields are in milliteslas.

ing point" of this regressive–transgressive megacycle of sedimentation occurs at the Ellis Bay – Becscie boundary and close to the inferred chronostratigraphic Ordovician–Silurian boundary (Petryk 1982). Evidence has been documented on the worldwide marine transgression in earliest Silurian time. An abrupt shift from a glacial climate to an interglacial one seems to be the most logical explanation for this phenomenon (Beuf *et al.* 1966; Berry and Boucot 1973; Harland and Herod 1975; Brenchley and Newall 1980). Although the explanation may be complex, there is convincing evidence to link changes in climate with changes in the Earth's magnetic field (Wollin *et al.* 1971).

Field procedure

The hand samples were collected (and oriented with a Brunton compass) by the junior author at 18 widely separate sites across the island (Fig. 1) and later cored in the laboratory. The number of samples per site varied from three to seven; the number of specimens per sample varied between two and five, for a total of 441 specimens. Most samples are from the Vauréal (six sites) and Ellis Bay (seven sites) formations. Fourteen samples are from the Becscie Formation, five are from the Gun River Formation, and five are from the Jupiter Formation. The geographic and stratigraphic locations (Fig. 1), relative ages, dip angles, and azimuths of the samples are presented in Appendix 1. The samples are listed in ascending stratigraphic order. The ages in parentheses refer to the informal North American, Anticosti Island stages and the European, British stages, respectively. Stage names are taken from Barnes and McCracken (1981).

Laboratory procedure

Magnetization (direction and intensity) was measured with a digital Schonstedt spinner magnetometer (DSM-1) coupled, in tandem, to PDP-11 and Apple II minicomputers. Two methods of demagnetization were used. Alternating field (AF) demagnetization was carried out using a solenoid having a maximum intensity of 90 mT, the performance of which was largely improved by adding three large, concentric mu-metal cylinders around the solenoid (Seguin 1975); a three-axis tumbler was used with the unit. Other AF demagnetizations were done using a Schonstedt GSD-1 apparatus (100 mT intensity). Thermal treatment of the specimens was done in a field-free oven (Schonstedt TSD-1). The amount of residual ambient field was 5 and 10 mT in the AF and thermal demagnetizers, respectively.

NRM directions and intensities

Site-mean natural remanent magnetization (NRM) directions (Fig. 2*a*), although characterized by a wide scatter of positively inclined steep remanence, have a slight deviation away from the present Earth's field (PEF) ($D = 335°$, $I = 75°$) at the site localities. The site-mean directions of the remanence are NNW and +50°, and the intensities vary from 10^{-4} to 5×10^{-3} A m^{-1}. A shallow inclination indicates that the geological formations carry some ancient remanence.

AF demagnetization

For the pilot AF study, at least five and up to 10 sedimentary specimens from each of the 18 sites were demagnetized in 7–13 steps up to 100 mT in some instances. For 90% of the samples, the intensities of magnetization fall off to less than 10% of the original NRM intensity in the interval 15–80 mT (average: 35 mT). For the same sedimentary units, the median destructive field (MDF) varies from 5 to 30 mT (average: 15 mT). In almost 30% of the samples, acquisition of spurious magnetizations at higher demagnetizing fields (40–100 mT) was observed. Generally, vector trajectories (Zijderveld 1967) are linear until stable end points are reached at higher fields, and an univectorial component is obtained once the low coercivity (soft component) is removed in the 2.5–15 mT range. Three components of magnetization are obtained: (A) NW, downward intermediate inclination, (B) SE, downward intermediate inclination, and (C) NW, upward shallow inclination. The coercivity spectra of these three components are quite similar. In all instances component A has a normal polarity and component C a reverse polarity; component B is mixed. Figure 3 and 4*a* show one typical example of the behaviour of these

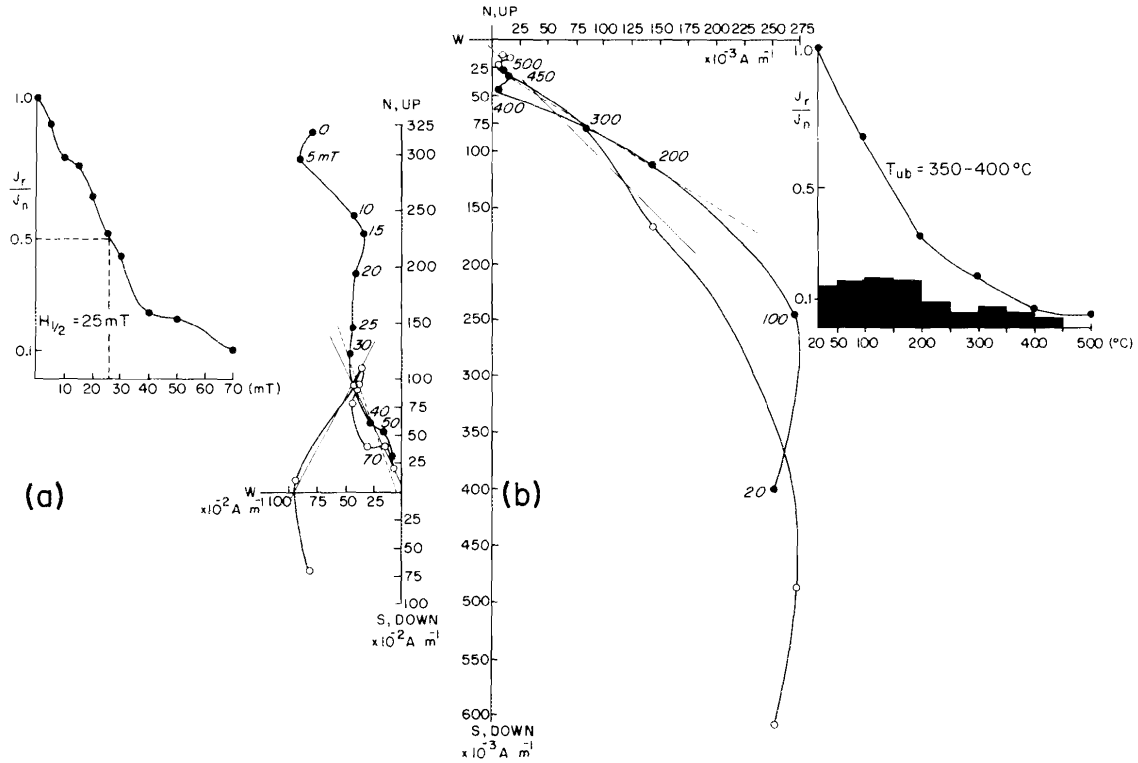


FIG. 4. Zijderveld (1967) diagrams showing the projection of magnetization vectors onto the horizontal plane (closed symbols) and the east–west vertical plane (open symbols). (*a*) Sample 7-2-3L-A shows a typical example of extraction of the B component (NNW, intermediate negative in polarity) by AF treatment. (*b*) Sample 13-2-1F-3 is a substandard to standard example of isolation of the same component (SSE, intermediate positive polarity) by thermal treatment. The median destructive field and coercitivity spectrum are shown in (*a*) and (*b*), respectively.

components upon AF demagnetization. Many of the cleaned directions are noisy. The directions of magnetization used for sample and site means were obtained using linear regression of the segment representing the higher coercivity region.

Thermal demagnetization

Six or more specimens from each of the 18 sites of the sedimentary units sampled were thermally demagnetized in a maximum of 10 steps up to 580°C, and vector diagrams (Zijderveld 1967) were used to analyse the results. Most commonly, the unblocking temperature (T_{ub}) is located in the 300–450°C range; occasionally it is at 500 and 525°C. In 90% of the samples, the intensities of magnetization fall off to less than 10% of the original NRM intensity in the interval 300–570°C (average: 450°C). Characteristic univectorial components are obtained through successive heating steps. The three components isolated in the course of AF demagnetization were also obtained by thermal treatment; an example of the behaviour of each of these components in the course of thermal demagnetization is presented in Figure 4*b* and 5. Components A, B, and C are characterized by N, M, and R polarities, respectively. In at least 75% of the cases, the AF and thermally demagnetized components in a sedimentary sample coincided. The isolated components are generally less noisy during thermal than during AF demagnetization. For this reason, about 65% of the specimens were thermally demagnetized. Low to intermediate coercivities, relativity low T_{ub} values, and optical microscopic observations (reflected light) indicate that both magnetite and titanomagnetite are the NRM carriers; hematite is definitely absent.

Statistical analysis of site-mean directions and relative age of magnetization

Site-mean directions for combined AF and thermally cleaned results are shown in Table 1 and Fig. 2*b* for both the Upper Ordovician and the Lower Silurian sedimentary units. A PEF component is relatively common at sites 1, 3–8, 12, 13, and 16–18. After AF and thermal demagnetization, three components of magnetization were isolated. Component A (NW, shallow to intermediate downward inclination) is predominant at sites 1–6, 9, 11, and 14. This component characterizes mainly the Vauréal Formation, as it is found at all the Vauréal sites. Except for site 14, the remaining A components are present in the Ellis Bay Formation. Component B (Fig. 4*a*) is predominant at sites 7, 8, 10, and 15; all these sites except site 15, belong to the Ellis Bay Formation. Component C is more common at sites 12, 13, and 16–18 (Fig. 4*b*), that is, in the upper part of the Ellis Bay, the Gun River, and Jupiter formations.

Component A is commonly found in the altered and indurated tops of samples from seven sites, and it is this component that was isolated in the preliminary paleomagnetic investigation (Seguin and Petryk 1984). In view of this observation, it is most probable that component A is secondary and of chemical origin. The second most common component encountered in other altered samples is B, which is found at five sites. Component C is rarely found in altered specimens. This last observation is an indication that component C is most probably the oldest component of magnetization. Component B is not a perfectly reversed direction of component C; thus these two components may constitute two different ones. Still, components B

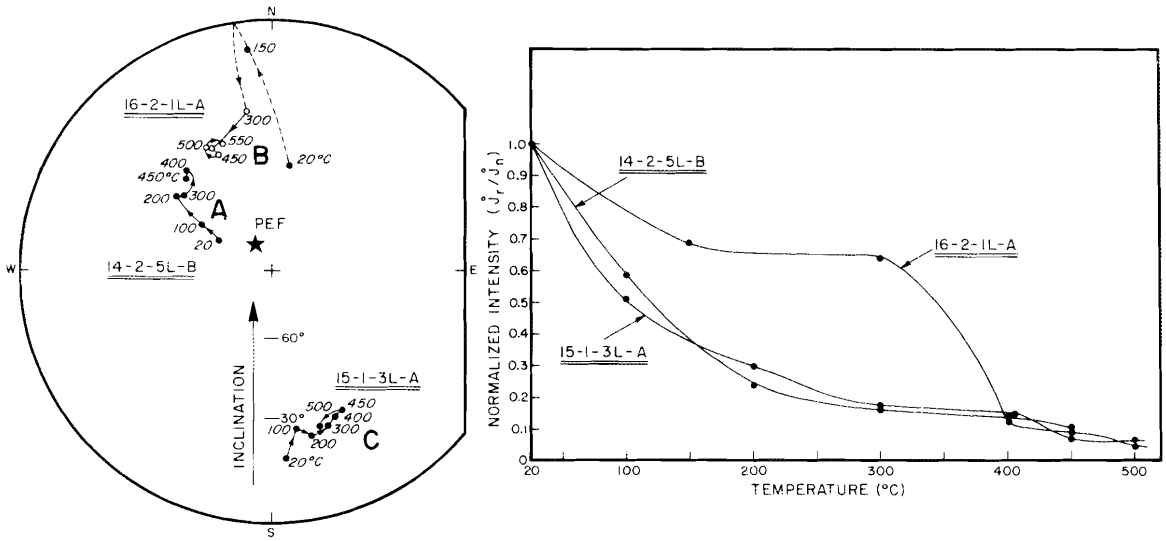


FIG. 5. Typical A, B, and C directions and intensities obtained by thermal demagnetization of specimens from low- and intermediate-intensity core. Intensities are normalized to the NRM values. Directions are plotted on a Wulff projection with closed (open) symbols indicating downward (upward) vectors. Temperatures are in degrees Celsius.

(reversed) and C are not significantly different at the 95% level of confidence and thus may represent the same episode of magnetization.

Tests based on stability criteria such as Graham's (1949) fold test cannot be performed to infer the age of magnetization of the Upper Ordovician to Lower Silurian lithological units because the beds have a uniform dip of 2°SW (Appendix 1). Contrary to the remanence of component A, which appears to be of secondary chemical origin, the remanence of component C carried in Ordovician–Silurian sediments could have a depositional origin. This conclusion is based on the presence of well-rounded, rather homogeneous and unaltered magnetite grains (high sphericity index), suggestive of penecontemporaneous deposition. Component B may be either primary or secondary. The nature of magnetite grains and the coherency in the direction of magnetization from formations of both Early Ordovician and Late Silurian ages may suggest a primary origin of detrital sedimentation (silty and sandy beds) within an overall predominant period of biogenic chemical (carbonate) deposition. Still, the arguments in favour of a primary origin of NRM in a structureless succession of rock lithologies that may have been and probably were remagnetized are rather weak.

Component C is overprinted by component A and is rarely found in association with component B. Components A and C are located on a subvertical plane going through the PEF (Fig. 6). This observation suggests that component A is the resultant of component C and the PEF. The vector-subtraction technique has confirmed this expectation (Table 2). The T_{ub} and H_c spectra of the B and C components are quite similar, and the sequence of acquisition of these components cannot be established on this basis.

Paleopole positions and their relation to poles from the North American craton

The paleopole position of component A (Fig. 7) is very close to the one obtained in a previous paleomagnetic study of the Vauréal and Ellis Bay formations (Seguin and Petryk 1984). Since component A is recognized to be a composite, no pole is quoted. The paleopole positions of the B and C components (Fig. 7) are 129°E, 19°N (dp = 12°, dm = 21°) and 161°E, 16°N (dp = 10°, dm = 18°). The pole of components B and C is noted in all formations. Assuming that the interpretation of a contemporaneity of component B and C is the correct one, these two components were then remagnetized in Silurian or later time, since they are found in the Silurian Becscie, Gun River, and Jupiter formations. When plotted on the proposed APWP for the North American craton, taking into account the existing Ordovician and Silurian paleopoles, the pole of component B falls close to the mean Early Silurian pole (S_1, Fig. 7), whereas component C falls close to the mean Early Ordovician pole (O_1, Fig. 7). This observation strongly suggests that the reference model of the early Paleozoic APWP for North America has yet to emerge.

Discussion and conclusion

Our analysis demonstrates that component A is predominantly found in the basal part of the lithological sequence studied; component B, in the middle part; and component C, in the upper part. This suggests a sequential process of magnetic overprints. In the basal part of the sequence, components A and C and the PEF are included in a subvertical plane oriented 315–320° (Fig. 6). Upon demagnetization (either AF or thermal), the magnetic vector generally moves from 320°, with intermediate to steep positive inclination, to 320°, with shallow positive or negative inclination. When the magnetic intensity becomes weak or the magnetization becomes unstable at higher AF strength or higher temperature, the inclination generally remains positive. When this is not the case, the inclination becomes negative. This indicates that declination and inclination of the end point of component C are close to 315°, −25° and that those of the end point of component B are close to 345°, −40° after reversing the direction. Vector subtraction of stepwise demagnetized values starting at D = 320°, I = 40° and ending at D = 315°, I = −25° indicates the occurrence of a steep northwest-oriented component that appears to be of recent origin. So component A is not really isolated and does not represent a unique component.

For components B and C, the following situations are pos-

TABLE 1. Statistics of cleaned components

Site	Formation	N'/N	n'/n	Component	D (°)	I (°)	α_{95} (°)	k
1	Vauréal	3/5	6/15	A	317	21	33	15
		2/5	4/15	B	174	52	—	—
		2/5	7/15	C	335	−04	—	—
2	Vauréal	3/3	9/9	A	321	43	29	18
3	Vauréal	2/2	4/7	A	305	45	—	—
		1/2	2/7	C	314	−25	—	—
4	Vauréal	6/6	9/16	A	304	28	25	14
		2/6	3/16	B	173	38	—	—
		1/6	3/16	C	326	−25	—	—
5	Vauréal	4/5	11/14	A	340	17	34	15
6	Vauréal	3/5	8/15	A	313	29	12	99
		2/5	5/15	B	160	41	—	—
7	Ellis Bay	2/4	6/10	B	147	41	—	—
8	Ellis Bay	2/5	5/12	A	302	43	—	—
		5/5	6/12	B	166	43	21	31
9	Ellis Bay	4/5	7/14	A	292	32	27	21
		2/5	5/14	B	202	23	—	—
		4/5	6/14	C	295	−31	22	29
10	Ellis Bay	6/6	11/13	B	158	38	28	20
11	Ellis Bay	3/3	6/7	A	321	35	19	17
12	Ellis Bay	4/5	11/13	C	301	−37	15	63
13	Ellis Bay	5/5	14/17	C	308	−28	14	43
14	Becscie	2/4	10/16	A	303	37	—	—
15	Becscie	2/5	6/13	A	298	43	—	—
		2/5	7/13	B	155	38	—	—
16	Becscie	2/5	5/17	A	354	36	—	—
		2/5	3/17	B	171	37	—	—
		2/5	6/17	C	307	−04	—	—
17	Gun River	2/5	3/16	B	184	06	—	—
		5/5	13/16	C	328	−29	28	23
18	Jupiter	2/5	3/15	B	144	49	—	—
		5/5	12/15	C	310	−27	33	9

Sites	Age	M'/M	Component	D (°)	I (°)	α_{95} (°)	k
1−11	Late Ordovician	9/11	A	313	33	09	29
	(O_L)	7/11	B	169	41	11	27
		4/11	C	319	−22	20	19
12−18	Early Silurian	3/7	A	318	40	33	16
	(S_E)	4/7	B	166	34	25	15
		5/7	C	311	−25	14	30
	O_L + S_E	12/18	A	315	36	21	—
		11/18	B(R)	347	−37	18	—
		9/18	C	315	−24	17	—
			B(R)+C	331	−30	—	—

NOTES: (R), reversed; N'/N, number of samples within this component over total number of samples; n'/n, number of specimens within this component over total number of specimens; M'/M, number of sites within this component over total number of sites; α_{95} and k, Fisher's circle of confidence and precision parameter, respectively, calculated using N', i.e., the sample as unit weight.

sible: (1) component B is older than component C, (2) component B is the same age as component C, or (3) component C is older than component B. In situation 1, component B may be primary or secondary; in situation 2, components B and C may be primary or secondary; and in situation 3, component C may be primary or secondary.

In view of the slightly higher T_{ub} and the smaller frequency of component C, as well as the relative absence of alteration on samples and sites containing component C, the third situation is more plausible than the first.

As components B and C are found in both Ordovician and Silurian formations, it is again plausible that they are secondary rather than primary. This means that the ages of acquisition of these two components are at least pre-Late Carboniferous, since they escaped the Kiaman remagnetization, and post-Early Silurian. If the smaller degree of alteration and the other magnetic characteristics mentioned above are significant, it can be concluded that component C is older than component B. If this argument is not retained, then B and C could be dual polarities of the same component, in which case the corresponding southerly paleolatitude represents a pre-Kiaman magnetization.

Assuming a secondary origin for components B and C, the Early Silurian − Early Carboniferous paleopoles for cratonic

TABLE 2. Examples of vector subtractions showing that component A is the resultant of component C and the PEF

Sample No.	AF demagnetization (mT)	D (°)	I (°)	J_r ($\times 10^{-3}$ A m^{-1})	Vector subtracted D (°)	Vector subtracted I (°)	Vector subtracted C
17-2-2L	0	316	20	3.25			
					292	12	
	2.5	335	18	2.92			
					312	78	PEF
	5	337	−3	2.58			
					282	29	
	10	297	−8	4.15			
					290	6	
	15	329	−18	3.28			
					347	−12	C
	20	330	−16	3.09			
17-2-3L	0	347	12	3.35			
					003	76	PEF
	5	346	1	3.10			
					332	55	PEF
	10	349	−19	2.69			
					303	38	
	15	354	−25	2.60			
					359	−17	
	20	349	−26	2.23			
					339	−6	C
	25	351	−31	1.80			
					346	−34	C
	30	347	−30	1.51			
					342	−36	C
	40	318	−17	0.79			
					269	−28	C
	50	326	−36	0.73			
					341	−42	C
	70	331	−29	0.35			

Sample No.	Thermal demagnetization (°C)	D (°)	I (°)	J_r ($\times 10^{-3}$ A m^{-1})	Vector subtracted D (°)	Vector subtracted I (°)	Vector subtracted C
3-1-2	20	345	5	7.43			
					315	65	PEF
	100	342	4	7.92			
					298	22	
	200	326	−12	12.20			
					322	10	
	300	327	−17	9.98			
					337	21	
	400	318	−22	9.43			
					331	−14	C
	450	322	−20	5.44			
9-12-2L	20	348	5	4.69			
					354	53	PEF
	100	346	3	4.08			
					333	22	
	200	344	−13	2.80			
					337	28	
	300	336	−24	2.44			
					345	−22	
	400	337	−26	1.20			
					337	−12	C
	450	325	−7	1.06			
					325	−21	C
	500	315	−29	0.18			

NOTE: C, component extracted.

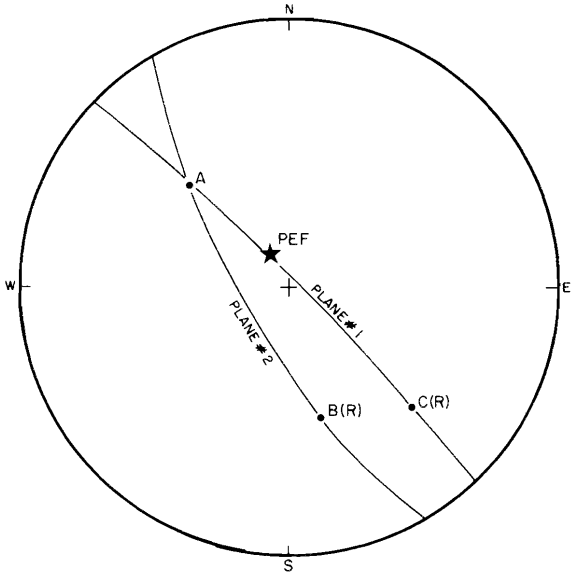


FIG. 6. Relationships of the A, B, C, and recent components of magnetization in space and interpretation of their time sequence in terms of vector subtraction.

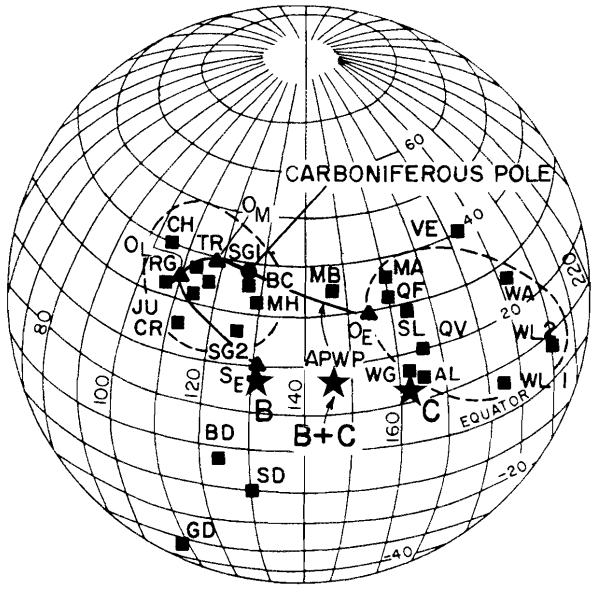


FIG. 7. Poles obtained from paleomagnetic studies of Ordovician rock units from eastern North America. Pole symbols are from Seguin and Petryk (1984). Poles WL_1 and WL_2 are from the Weedon–Gould lavas (Seguin 1980). The APWP indicated by the triangles is the best estimate for Early, Middle, and Late Ordovician (O_E, O_m, O_L) and Early Silurian (S_E) poles available. The stars show the pole positions of components B, C, and B + C combined (this study).

North America are located between 129°E, 19°N and 162°E, 13°N. The paleolatitudes of these components are 12–20°S of the paleoequator.

Only well-dated remanence acquisitions and associated primary poles can be used to define and calibrate the Ordovician–Silurian APWP. Obviously, the present paleomagnetic data from Anticosti do not fulfill this requirement. However, it must be realized that a well-defined remanence direction and associated pole may be of great value, even though remanence is not primary. Indeed, well-defined secondary or overprinted remanences can provide us with a record of the field at some time in the past (J. L. Roy, personal communication, 1985). As more of these secondary poles become available, a pattern describing an APWP might eventually emerge, especially when the ages of these poles can be confined to specific and relatively narrow time intervals.

In this respect it is worth noting that most of the so-called Middle and Late Ordovician poles are close to the Carboniferous pole (Fig. 7). This proximity of poles evokes the following questions. Are these Ordovician–Silurian poles well defined and correctly dated, or are they the result of the inadequate removal of a Carboniferous overprint? Both poles B and C lie to the south (19 and 16°N) of the Carboniferous pole (40°N). Since the APWP for cratonic North America from the Carboniferous to the present moves northwards (from 40 to 90°N), then both are pre-Carboniferous, and they indicate that the Late Ordovician – Early Silurian pole should lie south of 17°N and not north of it as defined by the presently accepted reference early Paleozoic APWP for North America. In summary, because of the omnipresence of overprints, a detailed magnetostratigraphic record of the lower Paleozoic sequence of the St. Lawrence Lowlands, in general, seems improbable.

What is the geological meaning of this sequence of magnetization? How can the magnetic overprinting be explained satisfactorily? It is first noted that the lower Paleozoic sedimentary sequence is found on a platform located close to the Grenvillian crystalline basement and relatively far from the Appalachian parautochthonous and allochthonous domains. Second, a glance at the distribution of the magnetic components as a function of the stratigraphy indicates that the number of components increases with the age of the stratigraphic sequence. The youngest components are more commonly found in the older rock sequences; the oldest, in the younger rock sequences. The most plausible explanation implies a low-temperature chemical remanent magnetization (CRM) process of acquisition of magnetization related to the porosity of sedimentary rock units and to alteration effects associated with downward-percolating or subhorizontally flowing water. As the amount of percolated or flowing water is probably proportional to the thickness of the stratigraphic column, the deeper lithological units are likely to contain a larger number of magnetic components. The reasoning behind this argument is that if all other parameters are kept constant, the older the sedimentary units are, the larger is the cumulated flow of subhorizontally to upward-circulating or percolating water. Admittedly, this phenomenon will change the magnetic properties of the opaque minerals, but it will not necessarily change their textural character. Ascending waters in the Anticosti sequence have probably been influenced by temperature gradients. Such thermally driven connate (original formation) waters have been periodically displaced by or mixed with descending meteoritic waters on a regional scale and over long periods of geological time. Consequently, it is sometimes impossible to tell whether the magnetization is of detrital (close to primary) or of subsequent CRM origin.

The above-mentioned general hydrologic model is not unusual for sedimentary sequences. However, as a corollary it must be remembered that in general, rock porosity decreases with depth, and so does the chemical equilibrium (fugacity) of oxygen. Both factors tend to diminish the influence of a secondary chemical remagnetization in rocks. The key to a

tighter constraint on the mode of acquisition of postdepositional remanence is related to a better understanding of later physicochemical processes affecting the original environment of deposition.

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Appendix 1: Geographic locations, stratigraphic locations, relative ages, dip angles, and azimuths of the samples

Site 1, samples 1 to 5

High shoreline outcrop, about 1.5 km before first major creek, west-southwest of Cape Rabast (Cap de Rabast) (Nord Point (Pointe Nord)), on the northwest coast; about 20.3 m from the base of the exposed Vauréal Formation (type section) on the island, in member 1; Late Ordovician (Richmondian: Cautleyan); dip 2°; 202°SW.

Site 2, samples 1 to 3

Wave-cut terrace outcrop (at low tide), about 0.5 km north of Ouest Point (Pointe de l'Ouest), westernmost tip of the island; about 400 m above the base of the exposed Vauréal Formation (type section), in upper member 3; Late Ordovician (Richmondian: Cautleyan); dip 2°; 202°SW.

Site 3, samples 2 to 5

Roughly the same location as site 2; about 2 m stratigraphically above site 2 samples, about 402 m above the base of the exposed Vauréal Formation (type section), in upper member 3; Late Ordovician (Richmondian: Cautleyan); dip 2°; 202°SW.

Site 4, samples 1 to 6

Upper shoreface outcrop, southwest of picnic area and roughly 1 km north-northwest of the abandoned barn, in the bight of Fraises Cove (Anse aux Fraises), on the west-southwest coast; about 424.5 m above the base of the exposed Vauréal Formation (type section), in upper member 4; Late Ordovician (Richmondian: Cautleyan); dip 2°; 180°S.

Site 5, samples 1 to 4

Southeast end of the shoreline outcrop at Batterie Point (Pointe à la Batterie), about 1.5 km east-southeast of the mouth of Saumons River (Rivière aux Saumons), on the east-northeast coast; roughly several metres above the site 4 stratigraphic level but below the reefal zone in upper member 4 of the Vauréal Formation; Late Ordovician (Richmondian: Cautleyan); dip 2°; 180°S.

Site 6, samples 1 to 4

Wave-cut terrace outcrop (at low tide) west of the abandoned barn in the bight of Fraises Cove, on the west-southwest coast; about the 432.2 m level above the base of the Vauréal Formation (type section), basal member 5; Late Ordovician (Richmondian: Cautleyan?); dip 2°; 180°S.

Site 7, samples 1 to 5

Cliff section at Cape Blanc (Cap Blanc), about 1 km southeast of the Port-Menier village wharf, off Ellis Bay (Baie Ellis), in the northwest part of the island; about 5 m below key bed 3 of member 3 of the Ellis Bay Formation; Late Ordovician (Gamachian: Rawtheyan?); dip 2.5°; 177°SE.

Site 8, samples 1 to 6

Below the major cliff section at pool 8 (rock bridge area) on Saumons River, in the southeast of the island; in uppermost nodular member 3, Ellis Bay Formation, between 15 and 35 cm below the member 3–4 contact at about river level; Late Ordovician (Gamachian: Rawtheyan?); 2°; 190°SW.

Site 9, samples 1 to 5

River bed or base of small section, about 400 m east of the bottom of the small branch of road leading down to pool 8 on Saumons River in the southeast part of the island; uppermost beds of nodular member 5 of the Ellis Bay Formation type section; Late Ordovician (Gamachian: Hirnantian A1?); dip 4°E; 10°NE.

Site 10, samples 1 to 7

Upper shoreface outcrop, below the bioherm-exposing cliff sections at Laframboise Point (Pointe Laframboise) on the west-southwest coast; the upper 13 cm of member 5 of the Ellis Bay Formation type section; Late Ordovician (Gamachian: Hirnantian A1?); dip 2.5°; 170°SE.

Site 11, samples 1 to 3

Upper shoreface outcrop in the area of site 10; stratigraphically in member 6 and about 13 cm below the basal oncolitic platform bed (OPB) of member 7 of the Ellis Bay Formation type section; Late Ordovician (Gamachian: Hirnantian A1?); dip 2.5°; 170°SE.

Site 12, samples 1 to 5

Undercut cliff section, in the area southeast of Ouest Cliff (Falaise Ouest), on the west shore of Ellis Bay in the western end of the island; all from the OPB of member 7 of the Ellis Bay Formation: measured from the top of the OPB, samples 1 to 5 are at 27, 27, 16, 14, and 25 cm, respectively; Late Ordovician (Gamachian: Hirnantian A1?); dip 2.5°; 180°S. Note that these samples are from the proposed site of the Ordovician–Silurian boundary stratotype.

Site 13, samples 1 to 5

In the same area as site 12, but 100 m southeast of that site; from the off-reef, reefal beds of member 7, Ellis Bay Formation, in the proposed Ordovician–Silurian boundary stratotype area: measured from the top of the OPB, samples 1 to 2 are at 80 cm and samples 3 to 5 are at 10 cm; latest Ordovician or earliest Silurian (Gamachian or stage C ?: Hirnantian A1 or Rhuddanian A2); dip 2.5°; 180°S. Note that these samples are in the proposed Ordovician–Silurian boundary zone.

Site 14, samples 1 to 3

In small outcrop section of the upper shoreface, roughly 1 km west-northwest of the mouth of Becs-Scie River (Rivière aux Becs-Scie), on the southwest coast of the northwest part of the island; about 89 m above the base of the Becscie Formation (type section), in member 2; Early Silurian (stage C; Idwian B1 or B2?); dip 1.5°; 182°SW.

Site 15, samples 1 to 5

River-level outcrop, at base of a 20 m section on the left bank, about 0.4 km south, downstream from Jupiter 24 bridge on Jupiter River (Rivière Jupiter) in the west-central part of the island; uppermost member 2 beds of the Becscie Formation: samples 1 and 2 are from the topmost and key rippled bed, samples 3 and 4 are from 20 cm below the top, and sample 5 is from 70 cm below the top of member 2; Early Silurian (stage B: Idwian B2?); dip 1.5°; 225°SW.

Site 16, samples 6 to 10

In same location as site 15, but about 30 m farther downstream; lowermost member 3 of the Becscie Formation, about 1.65 m above the member 2–3 contact; Early Silurian (Stage B; Idwian B2?); dip 1.5°; 255°SW.

Site 17, samples 1 to 5

River level, upstream end of east-bank outcrop on Jupiter River, about 140 m south of the confluence with Brise-Culotte Brook (Ruisseau Brise-Culotte), in the west-central part of the island; upper member 4 beds of the Gun River Formation, roughly 8 m below the Gun River – Jupiter contact; Early Silurian (stage B: Fronian C1?); dip 1–3°; 160°SE.

Site 18, samples 1 to 5

Undercut, 10 m sea-cliff section at "the Jumpers," about 300 m southeast of the south-southeast-bearing stretch of the road leading to the coast southeast of Sud-Ouest Point (Pointe du Sud-Ouest), in the south, west-central part of the island; uppermost metre of the beds of member 7 of the Jupiter Formation; Early Silurian (Stage A: Telychian C5?); dip 1.5–2°, 200°S. The samples were probably subjected to creep and rotations of 2–3°, opposite to mean dip.